A25 – Data Pipeline 1 – Project

This project focuses on designing and implementing an XML-based database platform for a **sports club**. The system will integrate and manage diverse entities—such as members, coaches, training sessions, facilities, teams, competitions, equipment, memberships, and bookings—to ensure efficient coordination and streamlined access to information. The project will cover:

- data modeling and management: Structuring and maintaining information about all entities and their relationships, including membership records, team organization, scheduling, and facility usage.
- user-oriented functionalities exploiting the database and supporting various user needs, such as searching for training sessions or teams, booking facilities or classes, managing memberships and renewals, etc.

The platform will showcase effective exploitation of XML technologies for real-world sports club operation and resource management.

- Write an XML Schema that represents your modeling
- Write an XML database that is valid against this model, containing a representative extract of the database.
- Imagine 6 different scenarios requiring the visualization of part of this data and/or new data computed from it and write 6 XSL stylesheets to implement them. (visualisation meaning outputting HTLM data)
- Imagine 2 additional scenarios requiring the exploitation of part of the XML database in another XML format and write an XSL stylesheet for this.
- Imagine 2 additional scenarios requiring the exploitation of part of the XML database in JSON format and write an XSL stylesheet for this.
- Write a Python program to load an XML file, parse it, validate it against the XML schema and apply an XSL stylesheet (using XML APIs).
- Optionally, write a Java program to load an XML file, parse it, validate it against the XML schema and apply an XSL stylesheet (using XML APIs).
- Optionally, provide a second implementation of one of your scenarios in Python or Java (using XML APIs, e.g. DOM).
- All the xml/xsd/xsl/json, java and python codes should be commented.

- Write a report (pdf format) containing the following:
    - indicate how you distributed the work load in the group (overall percentage of contribution to the project for each student and precise list of tasks to which each member contributed
    - indicate your working environment, and the tools or online services that you used.
    - describe the principles of your models and explain and discuss (advantages/disadvantages) your modeling choices. Choose one modeling problem that you adressed and explain the solution that you chose. **Nb: I do not expect here a general discussion on the XML language.**
    - describe the scenarios for which you implement a solution. Choose one scenario requiring quite a complex solution and explain it.
    - Additionnally, in case you make use of chatGPT, copilot or any other AI assistant at some points in the project, write an additional dedicated section where you indicate the precise prompts that you used, what you got, what you corrected, etc.
    - The report should not exceed 6 pages (optional section on AI assistance excluded).

You will be evaluated according to the ***quality and richness of your modeling*** (modular schema, fine grained type definitions and constraints), the ***quality and richness of the addressed scenarios/use cases/functionalities***, the ***quality of your stylesheets*** (prefer a recursive XSL programming style that extensively uses XPath), the quality of your Java and Python codes, and the quality of your report showing ***your ability to take a step back***.

Upload a **zip file** containing:

- your report (in pdf format)
- your source XML file
- your XML Schema file
- your XSLT files containing in comments their description in natural language
- the output of your XSLT transformations
- the JSON Schema of the 7th output
- your Python and Java source files